Week 2

Wednesday, February 4, 2026 7:15 PM

Module 4 : Gaussian Elimination

1. Write down the augmented matrix (remember it's in the form of [a1 a2 a3 | b]
2. Find the reduced echelon form of the matrix
    a. All non-zero rows must be above rows where there's all zeros
    b. The leading entry of the succeeding row must be strictly to the right of the previous row
    c. All leading entry must be equal to 1
    d. There's should only be one pivot variable in each pivot column (or each leading entry should be the only thing in its column)
3. Write down the equations corresponding to the reduced echelon form
4. Free variables are free, pivot variables has an equation

Theorem: Impossible to find solution when a row in a form of [0 0 0 0 0 | b], the system would not be system.

1. A unique solution (no free variables)
2. Infinitely many solutions (at least one free variable)
3. No solution (system is inconsistent)

Module 5: Linear Combination

Definition: If A is m x n, the transpose of A is the n x m matirx, denoted by AT, who columns are formed from the corresponding rows of A.

Definition: The linear combination of m x n matrices, A1, A2 .. Ap with coefficient c1, c2, … cp is definited as

C1A1 + C2A2...

Definition: The span (A1 … Ap) is defined as the set of all linear combination

How do I find the coefficient of the matrices?

1. Write them in the form of an equation
2. Put them into augmented matrix form
3. Turn into RREF
4. Check is system is consistent or not

Geometrically

- Span is the set of linear combinations that forms a plane

Module 6 – Matrix Vector Multiplication

Follow the form of

Ax = x1a1 + x2a2 + … +xnan

(x1, x2, … xn) is a solution of the system with augmented matrix
[A |b]

Now, introducing matrix machine.

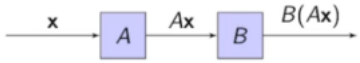
Suppose you have a matrix A and you want to know what this matrix A machine does.

First, let x be a column vector (it acts as an input).

Find the solution to Bx to get the output/result of the machine.

Composition of machine

Now, we're composing to machines.



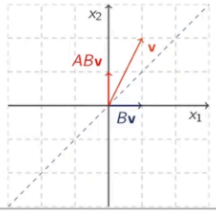
Note: If A is an m x n matrix and x in Rn, then Ax is in Rm. In order to calculate B(Ax) when then need B to have m columns.

Example. Let $A=\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}, B=\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}$ be as before. Is $A(B\mathbf{x}) = B(A\mathbf{x})$?

Solution.

No, projection and reflection do not commute!

$$A(B\begin{bmatrix}1\\2\end{bmatrix}) = A\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}0\\1\end{bmatrix}, \qquad B(A\begin{bmatrix}1\\2\end{bmatrix}) = B\begin{bmatrix}2\\1\end{bmatrix} = \begin{bmatrix}2\\0\end{bmatrix}$$



R means set of real numbers

Module 7 – Matrix Multiplication

To do matrix multiplication, you multiply each element of the row in the first matrix with the corresponding element of the column in the second matrix. You multiply each element together and add them together. This is like DOT PRODUCT.

Suppose we have matrix A and B. Note that:

Ab1 is a linear combination of the columns of A and Ab2 is a linear combination of the columns of A. Each column of Ab is a linear combination of the columns of A using coefficients from the corresponding columns of B.

Solution.

$$A\mathbf{b}_1 = \begin{bmatrix}4 & -2\\ 3 & -5\\ 0 & 1\end{bmatrix}\begin{bmatrix}2\\6\end{bmatrix} = 2\begin{bmatrix}4\\3\\0\end{bmatrix} + 6\begin{bmatrix}-2\\-5\\1\end{bmatrix} = \begin{bmatrix}-\\-\\6\end{bmatrix}$$

$$A\mathbf{b}_2 = \begin{bmatrix}4 & -2\\ 3 & -5\\ 0 & 1\end{bmatrix}\begin{bmatrix}-3\\-7\end{bmatrix} = -3\begin{bmatrix}4\\3\\0\end{bmatrix} - 7\begin{bmatrix}-2\\-5\\1\end{bmatrix} =$$

Matrix multiplication is only defined when the number of columns in A is the same the number of rows in B.

We already learnt that we can think of matrices as machines.

- Let B be $n \times p$: input $\mathbf{x} \in \mathbb{R}^p$, output $\mathbf{c} = B\mathbf{x} \in \mathbb{R}^n$.

Now, we're wondering if there is any correlation between matrix multiplication and composition of matrix.

$$A = \begin{bmatrix}2 & 0\\ 1 & 1\end{bmatrix}, \quad B = \begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

Compute $(AB)\mathbf{x}$ and $A(B(\mathbf{x}))$. Are these the same?

Solution.

$$B\mathbf{x} = \begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} = x_1\begin{bmatrix}1\\0\end{bmatrix} + x_2\begin{bmatrix}2\\1\end{bmatrix} = \begin{bmatrix}x_1 + 2x_2\\ x_2\end{bmatrix}$$

$$A(B\mathbf{x}) = \begin{bmatrix}2 & 0\\ 1 & 1\end{bmatrix}\begin{bmatrix}x_1 + 2x_2\\ x_2\end{bmatrix} = (x_1 + 2x_2)\begin{bmatrix}2\\1\end{bmatrix} + (x_2)\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}2x_1 + 4x_2\\ x_1 + 3x_2\end{bmatrix}$$

$$(AB)\mathbf{x} = \begin{bmatrix}A\mathbf{b}_1 & A\mathbf{b}_2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} = x_1 A\mathbf{b}_1 + x_2 A\mathbf{b}_2 = x_1(1\begin{bmatrix}2\\1\end{bmatrix} + 0\begin{bmatrix}0\\1\end{bmatrix}) + x_2(2\begin{bmatrix}2\\1\end{bmatrix} + 1\begin{bmatrix}0\\1\end{bmatrix})$$

$$= x_1\begin{bmatrix}2\\1\end{bmatrix} + x_2\begin{bmatrix}4\\3\end{bmatrix} = \begin{bmatrix}2x_1 + 4x_2\\ x_1 + 3x_2\end{bmatrix}$$

Theorem 1. *Let A be an $m \times n$ matrix and B be an $n \times p$ matrix. Then for every $\mathbf{x} \in \mathbb{R}^p$*

$$A(B\mathbf{x}) = (AB)\mathbf{x}.$$

Matrix Properties

(a) $A(BC) = (AB)C$ *(associative law of multiplication)*

(b) $A(B+C) = AB + AC$, $(B+C)A = BA + CA$ *(distributive laws)*

(c) $r(AB) = (rA)B = A(rB)$ *for every scalar* r,

(d) $A(rB + sC) = rAB + sAC$ *for every scalars* r, s *(linearity of matrix multiplication)*

(e) $I_m A = A = AI_n$ *(identity for matrix multiplication)*

Theorem 8.2. *The transpose of a product is the product of transposes* ***in opposite order****:*

$$(AB)^T = B^T A^T$$

Definition 8.2. Let A be a square matrix. We write A^k for $A \cdots A$, k-times; that is A^k is obtained by multiplying A k-times with itself.